



Investigation report

Grounding on Boolarong Bank, Weerona Channel, Kanooka Heads - Standard investigation

Drafted 30/09/2026

INVESTIGATION WORKFLOW SUITE

Contents

Executive summary	3
What happened	3
Outcome	3
Why it happened	3
Safety issues	4
Safety action	4
Event overview	5
Change of master and departure	5
Entering the Weerona Channel	5
The plotter freezes	5
Leaving the channel	6
The grounding	6
Immediate response	6
Outcome	7
Analysis	8
People	8
Environment	11
Equipment	13
Procedures	15
Organisation	16
Hypotheses tested and excluded	19
Findings summary	21
Contributing factors	21
Other factors that increased risk	21
Other findings	22
Appendices	23
Appendix A - Safety factor test summary	23
Appendix B - Excluded hypotheses	24
Appendix C - Sources of evidence	25
Appendix D - Investigation activity	26
Distribution	27

Executive summary

What happened

On the evening of 07/07/2026, the passenger ferry MV Tarrakin Star, a 24.4 m aluminium catamaran certificated for 96 passengers, was inbound through the Weerona Channel at Kanooka Heads on TF18, the day's last service. It carried 38 passengers, a master, a deckhand and a kiosk attendant, in darkness on a falling tide.

At 18:07:06 the chart plotter display froze and did not recover. The master continued at about 8 kt, searching visually for the No. 4 beacon. The heading increased by five degrees and was held, and the vessel set west out of the 40 m wide dredged channel. At 18:12:04 it struck Boolarong Bank at 6.8 kt, coming to rest 74 m west of the charted centreline and 54 m outside the dredged channel.

Outcome

There were no serious injuries. One passenger reported a grazed shin and two were treated at the wharf for distress. The port workboat had transferred all 38 passengers ashore by 19:26.

The vessel was substantially damaged:

- starboard hull bottom plating set up over 4.1 m, with a 180 mm split that flooded the No. 2 void to approximately 260 mm
- starboard propeller and shaft bent
- starboard rudder stock distorted
- starboard skeg torn away.

It refloated at 00:20 and was withdrawn from service. The Weerona Channel was closed to commercial traffic for 19 hours.

Why it happened

The plotter freeze removed the primary means of position fixing. The master continued at unchanged speed rather than stopping in the channel, which remained possible for the 114 seconds before the vessel reached the channel edge. This decision extremely likely contributed.

The operator had no night passage plan for the channel and no hold point for loss of the plotter, which very likely contributed. A winter timetable change had moved TF18 into darkness without reopening a route risk assessment that restricted the channel to daylight transits, which likely contributed.

Over the half mile before the vessel left the channel, mangrove canopy on Weerona Islet very likely screened the No. 4 beacon, the passage plan's secondary means of fixing. The Port Authority had deferred clearing that sightline, unfunded, at three works rounds, which very likely contributed. The beacon's solar conversion in 11/2025 had also halved its nominal range. No operator was told, and the passage plan still recorded the old range. This likely contributed, although the light was screened where the master was looking.

The plotter fault had occurred twice that morning. It was passed on verbally at the change of master but not entered in the defect book, which likely contributed. A firmware fix issued in 11/2024 had never been applied.

The echo sounder shallow alarm, at its 2019 commissioning setting of 0.9 m, gave 16 seconds' warning, too little time to stop. This extremely likely contributed.

Safety issues

Six contributing safety issues were identified:

- the timetable change process did not reopen the route risk assessment
- there was no night passage plan and no hold point
- the change-of-master defect brief is verbal, with no check that what is said is written down
- the channel is not a declared shipping channel, so its aids to navigation carry no availability standard and no duty to notify users
- the solar conversion reduced the beacon's range without notice to operators
- clearing of the beacon's sightline was deferred.

The influence of the defect brief rule and of the waterway standard was argued rather than settled. For each, the evidence also supports a reading as a genuine safety concern that was not a cause of this grounding.

A seventh safety issue was examined and not established as contributing: the forward saloon lifejacket locker cannot be reached from most of the saloon it serves. The arrangement dates from a 2019 refit that did not trigger a re-survey, and the same counter is fitted to two other vessels. The master's split-duty roster also increased risk, but was not shown to have contributed.

The investigation could not determine whether new foreshore floodlighting behind the beacon affected the master's ability to see it.

After the grounding, the master's decision not to attempt to refloat on a falling tide very likely reduced the risk to those on board.

Safety action

[NOT ESTABLISHED: safety actions taken by the operator, the Port Authority or the Council, and safety actions recommended]

Event overview

On the evening of 07/07/2026, MV Tarrakin Star, a 24.4 m aluminium catamaran passenger ferry certificated for 96 passengers, was operating service TF18, the last of the day from Muntari Island to the town wharf at Kanooka Heads. All times are local (UTC + 10 hours).

Since 01/06/2026, TF18 and the winter timetable's other evening returns had been made in darkness. Sunset was at 17:02 and civil twilight ended at 17:29. The master described the conditions as clear, with a light westerly and no sea in the entrance.

The crew were the master, a deckhand and a kiosk attendant. The master had 14 years as a master, 11 on this route, and about 4,100 Weerona Channel transits. He was on day nine of a twelve-day tour, working a split duty. The deckhand had been with the operator for 7 months and was in her first winter season on the route.

Change of master and departure

The master relieved the morning master at the town wharf at about 13:05. The chart plotter's event log recorded display freezes at 10:42 and 11:17 that morning, each followed by an automatic reset. Both masters' accounts were that the outgoing master mentioned the plotter had frozen twice and recovered by itself, and the relieving master replied that it had done the same to him the month before. Neither entered it in the wheelhouse defect book, which held no entry of any kind for 07/07/2026 before the occurrence.

The afternoon's two round trips were unremarkable. At 17:32 the master told passengers:

- "we're about twenty five minutes across to the town wharf tonight."

TF18 departed at 17:35 with 38 passengers. The master reported the departure to Port Control on VHF channel 12, which replied:

- "nothing to affect you, channel's clear."

Entering the Weerona Channel

The vessel entered the channel from the north and was abeam the No. 1 beacon at 18:03:58, 2 m east of the charted centreline, making 8.4 kt over the ground on 195 degrees. At 18:04 the deckhand asked whether she should stay on the bridge. The master replied:

- "Nah, go and do the fares, I'll give you a yell coming up on the wharf."

The deckhand was in the saloons from about 18:05, leaving the master alone on the bridge for the rest of the transit. At 18:05:30 the vessel passed the No. 2 beacon 1 m west of the centreline, still at 8.4 kt on 195 degrees. The master remarked:

- "Two ... good."

The plotter freezes

At 18:07:06 the plotter's event log recorded that the display process had stopped responding; unlike the morning events, no reset followed. The master's account was that the chart stayed on screen with the vessel symbol stationary and the display did not respond when he tried it twice. The bridge audio captured "Oh, come on" at 18:07:11, sounds consistent with repeated screen taps, and at 18:07:44 a partly unintelligible remark ending "frozen again".

The master continued at unchanged speed, navigating by eye and looking for the No. 4 starboard-hand beacon on the western side of the channel, on a transit he described as one he had made thousands of times.

Leaving the channel

From 18:07:06 the heading increased steadily from 195 to 200 degrees; the rate of turn peaked at 2.1 degrees per minute at 18:07:40 and returned to zero by 18:08:50, after which the new heading was held. The vessel set west at an average of about 12 m per minute. The master's account was that he eased five degrees to starboard while searching for the beacon, without deciding to do so.

Local time	Position west of centreline	Distance north of No. 4 beacon	Speed	Event
18:08:31	12 m	0.47 nm	8.0 kt	Master says "Right. Where's four."
18:09:00	20 m	0.40 nm	8.0 kt	At the channel's western edge
18:09:40	—	—	—	Sound consistent with a window or door being slid open
18:10:02	39 m	0.25 nm	7.9 kt	Master says "There's four. Keep her over."; outside the dredged channel
18:11:00	58 m	0.13 nm	7.6 kt	Over Boolarong Bank

The master's account was that he opened the wheelhouse window in case the glass was the problem. He sighted the beacon 91 seconds after he began looking; his account was that it was dead ahead rather than on the starboard bow, and he understood at once he was on the bank side of the channel.

The engine monitoring recorder showed both engines steady at 1,780 and 1,784 rpm, with no throttle movement between 18:03 and 18:11:52. Speed over the ground fell from 8.4 kt to 7.6 kt between 18:07 and 18:11.

The grounding

At 18:11:48 the echo sounder's shallow alarm, set at 0.9 m under the keel since commissioning in 2019, sounded with the vessel 70 m west of the centreline at 7.1 kt. The master said:

- "No no no."

At 18:11:52 both throttles were moved to astern together. Before astern thrust reached full, at 18:12:04, the vessel struck Boolarong Bank at 6.8 kt. The alarm had given 16 seconds' warning, in which the vessel covered 58 m; the trials figure for a crash stop from 8 kt is 92 m.

The bridge audio recorded a heavy impact followed by grinding and vibration for approximately 9 seconds. The vessel stopped within 11 m, 74 m west of the charted centreline, 54 m west of the channel's western edge and 34 m west of the No. 4 beacon, where the bed was 0.50 m below chart datum. Furrows consistent with the starboard hull and skeg ran 11 m from first contact on a heading of 197 degrees.

The occurrence notification recorded the vessel as heeled 4 degrees to starboard and hard aground on a falling tide; in his broadcast shortly afterwards, the master described it as stable and upright.

Immediate response

The master stopped both engines at 18:12:31 and did not attempt to refloat, later citing the falling tide, the unknown bottom, the 38 people on board and the grinding felt on contact. At 18:12:40 he told passengers:

- "We have touched the bottom. The vessel is stopped and she is safe."

At 18:13:04 he broadcast a pan pan to Kanooka Heads Port Control on VHF channel 16, giving the vessel's position aground on Boolarong Bank abeam the No. 4 beacon and the persons on board, reporting no injuries or pollution, stating he was not attempting to refloat, and requesting the

workboat. Port Control acknowledged at 18:13:29 and asked whether the vessel was taking water. The master replied that the voids were being sounded and added:

- "Be aware the tide is falling, she will sit down further."

At 18:14:52 he asked passengers to put on lifejackets, telling them:

- "We are not going into the water."

At about 18:16 the deckhand reported about 75 mm of water in the No. 2 void; the master directed soundings every ten minutes and at 18:17:05 reported to Port Control slow ingress into the No. 2 starboard void and none elsewhere.

The deckhand issued lifejackets from aft forward, and all passengers had one by about 18:22. The aft saloon locker is on the bulkhead; the forward saloon locker is behind the kiosk servery counter, where three passengers were standing, so the deckhand had to go through a narrow gap at the end of the counter and hand jackets out over it.

The port workboat came alongside the port quarter at 18:41. At 19:26, 74 minutes after contact, the master reported all 38 passengers ashore, with the three crew remaining on board.

Outcome

There were no serious injuries. One passenger reported a grazed shin and two were treated at the wharf for distress. The vessel was refloated on the rising tide at 00:20 on 08/07/2026 and towed to the town wharf, by which time the No. 2 void held approximately 260 mm of water.

Component	Damage
Starboard hull	Bottom plating set up over 4.1 m between frames 6 and 11, with a 180 mm split in way of the No. 2 void
Starboard propeller and shaft	All three blades bent aft; shaft bent 1.4 mm
Starboard rudder stock	Distorted 3 degrees
Starboard skeg	Torn away
Port hull, propulsion and steering	No damage found

The vessel was withdrawn from service, and the Weerona Channel was closed to commercial traffic for 19 hours.

Analysis

This section analyses the conditions, actions and decisions that led to MV Tarrakin Star grounding on Boolarong Bank, and those that shaped what happened afterwards. It is organised under five headings: people, environment, equipment, procedures and organisation.

Each thread ends with the resulting finding. The strength of each finding is expressed on a fixed verbal scale, so that one finding can be compared with another. Contributing safety factors are distinguished from other factors that increased risk. One matter could not be resolved and is discussed as such.

People

Departure from the channel

The recorded track shows the vessel on the charted centreline until 18:07:06. It was 2 m east of the centreline at the No. 1 beacon and 1 m west at the No. 2 beacon.

From 18:07:06 the heading increased steadily from 195 to 200 degrees and the vessel set west at an average of about 12 m per minute. It crossed the western edge of the dredged channel at about 18:09 and was 39 m west of the centreline at 18:10:02. It was over Boolarong Bank from about 18:11 and came to rest 74 m west of the centreline, 54 m outside the channel.

The shape of the turn matters. The recorded rate of turn peaked at 2.1 degrees per minute at 18:07:40 and returned to zero by 18:08:50. The heading of about 199 to 200 degrees was then held. This is a heading change applied over about a minute and then kept, which is consistent with a vessel steered onto a new heading and held there. It is not a drift under current or wind:

- the wind was 305 to 310 degrees at 7 to 9 kt;
- the sea in the channel was slight, with wind ripple only; and
- there is no cross-set in the channel at that state of tide.

The deviation began within seconds of the plotter freezing and continued through the period in which the master was searching for the No. 4 beacon. The surveyed grounding position and the 11 m of furrows in the bed, on a heading of 197 degrees, corroborate the track.

The channel is 40 m wide against a vessel beam of 8.2 m and is dredged to 3.2 m below datum. Once the vessel was west of the dredged edge, the drying bank lay ahead on its track.

It is virtually certain that the vessel left the dredged channel and tracked onto Boolarong Bank, and virtually certain that this contributed to the grounding.

Continuing the transit after the plotter froze

When the plotter froze at 18:07:06, the vessel was 0.68 nm north of the No. 4 beacon, 3 m west of the centreline and making 8.4 kt.

The engine monitoring recorder shows no throttle movement from 18:03 until 18:11:52. Speed over the ground held at 8.4 kt to about 18:08 and fell to 7.6 kt by 18:11. That reduction is accounted for by the increasing drag of shoaling water beneath the hulls, not by any action on the bridge.

The bridge audio places the master's attention on the frozen display between 18:07:11 and 18:07:44. At 18:08:31, 85 seconds after the freeze, he said:

- "Right. Where's four."

Nothing on the audio indicates that stopping was considered at any point.

The conditions did not argue against stopping:

- there was no other traffic in the channel;
- the water was flat and the wind light;
- the vessel could have been stopped in 92 m, its trials figure for a crash stop from 8 kt; and
- 114 seconds passed between the freeze and the vessel crossing the channel's western edge.

A stop at any point in that window would have left the vessel in the channel rather than on the bank.

The master's account, in interview and in his safety report, was consistent. He did not consider stopping. He went straight to the visual transit because, in his understanding, the visual transit was the backup to the plotter. He said that had he been asked beforehand, he would have said he would stop.

That account is significant for how this action is understood. The operator's passage plan named the No. 4 beacon as the secondary means of position fixing. It said nothing about losing the primary means and named no point at which to hold. The master turned to the means the plan gave him. As discussed under Environment, that means was unavailable over the whole stretch in which he looked for it.

There was also no second person on the bridge. At 18:04:14, as the vessel entered the channel, the deckhand asked whether to stay up and was sent to collect fares. This is the normal practice on the route, and nothing in OPS-3 addresses it either way. With the master alone, the search for the beacon, the attempt to recover the plotter and the conduct of the vessel all fell to one person.

This action should be read in the context of the conditions that produced it. It sits downstream of a passage plan with no hold point and a beacon that could not be seen. Both are analysed below.

It is virtually certain that the transit continued at about 8 kt after the plotter froze, rather than the vessel being stopped in the channel, and extremely likely that this contributed to the grounding.

The handover of the plotter fault

The plotter's event log recorded two display freezes with automatic resets on the morning shift, at 10:42 and 11:17.

At the change of master at about 13:05 at the town wharf, the outgoing master told the relieving master about both. The relieving master replied that the unit had done the same to him the month before. Neither master entered the fault in the wheelhouse defect book or reported it to the Operations Manager, and the book held no entry of any kind for 07/07/2026 before the occurrence. The two masters' accounts of the handover agreed in every particular, and both volunteered the omission before being asked about it.

The omission was not isolated. The outgoing master believed the plotter had frozen three or four times on this vessel in the past year, to him and to others, and that none of those events had been written up. The maintenance system held no work order, fault report or correspondence about the plotter at any time in 2026. The operator held no copy of the manufacturer's bulletin and had no subscription to the manufacturer's bulletins. As a result, the firmware issued in 11/2024 to address cause code 0x51 had never been applied to this unit.

The case that the omission influenced the grounding rests on that wider pattern rather than on the single handover. An entry would have made the repeated freezes visible to the people who manage the vessel's equipment and decide whether it sails. The fault the master met at 18:07 was one the manufacturer had already addressed.

There is a strong counterpoint. A single entry made at about 13:05, reviewed on the weekly cycle that OPS-9 provides, would not by itself have stopped TF18 sailing that evening. For that reason the connection to this grounding is not certain.

The finding concerns the practice, not the two masters. Both described the same understanding of how defects are handed over. The manual, discussed under Procedures, bears that understanding out. This explains how the omission came about; it does not remove the omission as a link in the sequence.

It is virtually certain that the plotter fault was passed on verbally at the change of master and not entered in the defect book, and likely that this contributed to the grounding.

Duty and fatigue

The master's duty was examined as a possible influence on his decision-making.

The route is worked 12 days on and 2 days off, and 07/07/2026 was the master's ninth consecutive day. His duty was a split, 05:40 to 08:40 and 13:10 to 19:20. This gave 9 hours 10 minutes of duty across a spread of 13 hours 40 minutes, with an unpaid break of 4 hours 30 minutes at home in the middle. Both figures were inside the operator's own limits of 10 hours of duty and 14 hours of spread, and inside the state's crewing determination.

The master recorded 6 hours 15 minutes of sleep the night before in the fatigue diary. The operator asks masters to keep that diary but does not collect it. In interview he described himself as tired, said this was normal for a split duty, and said he knew where he stood on a roster that had not changed in eleven years.

The case for influence, put at its strongest, is this. Day nine of twelve on a split duty, after six and a quarter hours of sleep, falls in a range where degraded decision-making is expected. The decision that was not made, to stop when the plotter froze, is the kind of decision fatigue degrades. The pattern is also a standing one: a ninth consecutive day ending with a night channel transit is the ordinary shape of this roster in winter, and every master on the route works it.

The recorded evidence runs the other way:

- the master held the vessel within 3 m of the centreline for the first two minutes in the channel;
- he responded to the shallow alarm in 4 seconds, against a fleet figure of 6 to 9 seconds;
- his radio work after the grounding was complete and correct at the first attempt; and
- his decision not to refloat was made in about thirty seconds and was sound.

The one decision fatigue would have had to act through is the failure to consider stopping. Both masters said independently that nothing in any document asks a master to consider it. The outgoing master, who was not tired, described the same understanding of the handover and the same difficulty with the beacon. A decision that nobody was prompted to make cannot be shown to have been degraded by fatigue.

The factor therefore did not contribute to this grounding, but it remains a standing exposure:

- the winter timetable places the last darkness transit of the day at the end of a split duty whose spread comes within twenty minutes of the operator's limit;
- that duty falls on the ninth to twelfth consecutive day;
- the fatigue diary is requested but not read; and
- no relief master is rostered on the route, so a single unfit crew member extends a tour rather than breaking it.

It is virtually certain that the master was on the ninth consecutive day of a split duty with a spread of 13 hours 40 minutes; it is unlikely that this contributed to the grounding, and likely that it increased risk.

Actions after the grounding

The master's actions after contact reduced the consequences of the grounding.

He stopped both engines 27 seconds after contact and did not restart them. He addressed the passengers 36 seconds after the impact. In his pan pan at 18:13:04 he stated his decision as already made:

- "I am not attempting to refloat."

He had the No. 2 starboard void sounded, reported about 75 mm of ingress to Port Control at 18:17:05, and had the void sounded every ten minutes after that. He asked every passenger to put on a lifejacket. His reasoning in interview was that the tide was falling, the bottom was unknown, 38 people were aboard, and the vessel could not sink while it sat on the bank.

The survey later found a 180 mm split in the starboard hull plating in way of that void. Ingress had reached approximately 260 mm by the time the vessel refloated at 00:20. The breach was in the part of the hull resting on the bank. Backing the vessel off into 3.2 m of water, at night, on a falling tide, with the ingress unassessed and passengers aboard, would have put a flooding compartment into deep water.

It is virtually certain that the master stopped the engines and made no attempt to refloat on the falling tide, and very likely that this reduced the risk to those on board.

Environment

Darkness and conditions

TF18 departed at 17:35. The transit of the Weerona Channel began at about 18:04, some 35 minutes after the end of civil twilight at 17:29. The moon was a waning crescent, 2 per cent illuminated, which had risen at 06:12. Every reference available to the master for fixing position by eye was therefore an artificial light.

Visibility was greater than 10 km, cloud was few to scattered at 3,500 ft, no rain had fallen, and the sea in the channel was slight. These were not conditions of reduced visibility. That distinction matters for how the operator's procedures applied, which is discussed under Procedures.

The No. 4 beacon was screened from the vessel

At the investigator's request, a range and visibility check was made from the port workboat on 18/07/2026. It was carried out on the inbound track at the same time of night. It found that the No. 4 beacon cannot be seen at all from the channel between 0.84 nm and 0.34 nm north of it.

Over that stretch, the sightline passes through the mangrove canopy on Weerona Islet, 240 m north-west of the beacon. The canopy measured 5.8 m against a lantern height of 4.2 m above chart datum. The light is visible beyond 0.84 nm, where the sightline clears the islet's northern edge, and again inside 0.34 nm, where it opens past the southern edge.

The recorded track places the master's whole search inside that arc:

- the vessel was 0.68 nm north of the beacon when the plotter froze;
- it was 0.47 nm north when the master said "Right. Where's four." at 18:08:31; and
- it was 0.25 nm north at 18:10:02, when he said:
- "There's four. Keep her over."

The 91 seconds between those two remarks match the measured screening, and in that time the vessel crossed the channel's western edge. Both masters interviewed raised the beacon unprompted, and neither had ever reported it.

With the plotter frozen, the beacon was the only means of fixing position that remained. The town wharf leading lights, the other half of the plan's secondary means, do not resolve as two lights until the vessel is inside the southern third of the channel. The beacon was unavailable for the whole period in which the vessel set west. The master was not failing to see a light that was there. Over that half mile there was no light to see.

Two limits on this conclusion are recognised. First, the measurement was made eleven days after the occurrence, and the canopy was not surveyed on the night itself. Second, the vessel had already begun to alter heading before the master started looking for the beacon, so the screening did not by itself start the deviation. It did remove the one means by which the deviation might have been detected.

It is extremely likely that the No. 4 beacon was not visible from the channel over the half mile before the vessel left it, and very likely that this contributed to the grounding.

Foreshore floodlighting behind the beacon: an unresolved matter

The investigation examined whether new floodlighting on the Kanooka Heads foreshore, behind the No. 4 beacon as seen from the channel, reduced the master's ability to pick the beacon up. This question could not be resolved.

The evidence for an effect

- **Condition of the lighting.** The foreshore redevelopment was commissioned on 22/05/2026, ten days before the winter timetable began. It includes 14 LED floodlights on 12 m columns lighting the new boat ramp and car park. The survey on 18/07/2026 found that each presents about 4,800 candela toward the channel and none is shielded to seaward.
- **Alignment with the beacon.** From 0.5 nm north of the beacon, the array lies behind it within 4 degrees of the same bearing and within 1.5 degrees of the same elevation. The array is lit continuously, while the beacon shows a green flash of 0.4 seconds every 2.5 seconds.
- **The masters' accounts.** The master described looking into a wall of light, with the beacon amounting to nothing against the town and the new car park. The outgoing master, independently, described the beacon as washed out altogether from the northern approach.
- **Absence of assessment.** The development was approved by the Council's planning unit. The Port Authority's aids to navigation officer was not consulted, and no assessment of the lighting against the channel's aids was required or made.

The evidence against an effect

- **The screening.** Over the stretch where the master was looking, the beacon was hidden by the mangrove canopy. It would not have been visible against a dark background either.
- **Where the alignment was measured.** The one position at which the alignment was measured, 0.5 nm north, lies inside the screened arc.
- **The beacon was sighted with the floodlights unchanged.** The light opens clear of the islet from 0.34 nm, and the master called it at 0.25 nm with the floodlights in the same place and no dimmer.

- **No measurement of detection.** There is no measurement of what could be seen on the night, only the survey eleven days later. Nothing establishes a detection threshold for a 0.4 second green flash against that background.

Why the matter could not be resolved

The alignment is real, and the masters' accounts are consistent and independent. However, the evidence cannot separate the effect of the floodlights from the effect of the canopy. While the canopy screens the beacon along the stretch that mattered, the question may not be answerable at all.

Settling it would require an assessment of how detectable the beacon's flash is against the floodlit background from the inbound track, with the sightline clear. The investigation therefore draws no conclusion either way on whether the floodlighting affected the master's ability to see the beacon, and the matter is not carried into the findings.

The fact that a large, unshielded light source was placed behind an aid to navigation without any assessment against the channel's aids stands independently of this question.

Equipment

The chart plotter

The plotter is a 12-inch multifunction display fed by a separate GPS antenna unit. It was running chart software revision 4.11 on firmware issued in 03/2024.

Its internal event log recorded watchdog resets with cause code 0x51, meaning the display process was not responding, at 10:42 and 11:17 that morning. On each occasion the unit recovered itself. At 18:07:06 the log recorded the same condition with no reset, and the unit did not recover. The bridge audio matches the log to the second: "Oh, come on" at 18:07:11, sounds consistent with repeated screen taps at 18:07:26, and a partly unintelligible remark ending "frozen again" at 18:07:44. The master's account was that the chart stayed on screen with the vessel symbol stationary and the display not accepting touch input.

An approved electronics contractor bench tested the unit for 46 hours on 21/07/2026 with the same antenna and cabling. The fault did not recur and no defect was found. The contractor identified cause code 0x51 as a condition known to the manufacturer and addressed by firmware issued in 11/2024, which had not been applied to this unit.

This finding carries a qualification. Because the fault could not be reproduced and no physical defect was found, the evidence of the evening's failure is the software's own report of itself, supported by the audio and the master's account. It is not an independently demonstrated failure.

The influence of the freeze is clearer. The vessel held the centreline while the plotter worked and left it within two minutes of the freeze. From that point the transit was no longer the one described in the passage plan, because the master no longer had the primary means of position fixing the plan told him to use.

It is very likely that the chart plotter display froze during the channel transit and did not recover, and extremely likely that this contributed to the grounding.

The echo sounder shallow alarm

The echo sounder's shallow alarm was the last defence between the vessel and the bank. The alarm log states:

- "Shallow alarm set to 0.9 m under the keel. This is the value written into the setup when the unit was commissioned in 2019 and it has not been changed. The alarm sounded once on the voyage, at 08:11:48 UTC, 16 seconds before contact."

The log's time of 08:11:48 UTC corresponds to 18:11:48 local. The bridge audio records the tone beginning at that second, and both throttles were moved to astern together 4 seconds later. Astern thrust was developing but had not reached full when the vessel made contact at 18:12:04.

The alarm log explains why the warning was so short:

- "At 7.1 kt over the ground the vessel covered 58 m in those 16 seconds. At the surveyed bank gradient of about 1 in 70 across that section the alarm could not have sounded appreciably earlier at that setting. The vessel's own trials figure for a crash stop from 8 kt is 92 m."

The equipment did exactly what it was set to do. The shortfall lay in the combination of the setting, the gradient of the bank and the transit speed. A 0.9 m setting on a gradient of about 1 in 70 at about 8 kt provides less warning than the distance the vessel needs to stop. A setting that gave materially more warning, or a speed at which 16 seconds was enough to stop, would have left the vessel short of the bank.

It is virtually certain that the shoaling was not detected until 16 seconds before contact, and extremely likely that this contributed to the grounding.

Forward saloon lifejacket stowage (Safety Issue)

While issuing lifejackets after the grounding, the deckhand had to deal with the forward saloon locker, which is behind the kiosk servery counter where three passengers were standing. The deckhand asked the three passengers to move and passed behind the counter through a 0.6 m gap to reach it. At 18:22:40 the deckhand reported:

- "There's two in the forward saloon I had to go round the servery for, the locker's behind the counter and there's people standing at it."

A stowage survey was carried out on 16/07/2026, following the deckhand's account and safety report. It found the following:

- **Reach from the seats.** The forward saloon locker holds 40 adult lifejackets and is mounted in the bulkhead behind the servery counter. Its face cannot be reached from 22 of the 34 forward saloon seats without passing behind the counter.
- **Route with people at the counter.** With people standing at the counter, as on the night, the gap at the counter's inboard end is the only route to the locker.
- **Muster placard.** The placard directs passengers to that locker, which cannot be seen from the seats.
- **Origin of the arrangement.** The arrangement approved at build in 2011 had an open bench in that position, with the locker face clear. The counter was fitted during a 2019 interior refit under a minor modification notice, which did not trigger a re-survey of the safety equipment arrangement.
- **Other vessels.** The same counter is fitted to the operator's two other 24 m vessels.

This arrangement had no effect on the grounding and no material effect on its outcome. The vessel was aground, upright and stable. All 38 passengers were in lifejackets within about eight minutes of the master's request and ashore within 74 minutes of contact. The muster took a little longer than it should have, but it was not hurried.

The arrangement matters for other circumstances. In a fire, a risk of capsize, an evacuation into the water at night, or with a crew member injured in an impact, a locker that passengers cannot reach without crew help would decide whether the forward saloon can equip itself or must wait.

The underlying gap is the modification route rather than the counter itself. An interior refit changed a safety equipment arrangement, and nothing in that route asked whether it had done so. The arrangement is repeated across the operator's vessels of this size.

It is virtually certain that the forward saloon lifejacket locker could not be reached from most of the saloon it serves; it is unlikely that this contributed to the grounding or its outcome, and very likely that it increased risk.

Procedures

No night passage plan, and no hold point if the primary means of fixing is lost

The transit was conducted under Tarrakin Ferries Operations Manual OPS-3. OPS-3 section 3.4 states:

- "The Weerona Channel transit will be made on the charted centreline, monitored on the chart plotter, with the No. 2 and No. 4 beacons identified visually before the vessel passes the No. 1 beacon."

The Weerona Channel passage plan, OPS-3 Appendix A issue 2, dated 02/2023 and current, specifies:

- "Primary means of position fixing: chart plotter. Secondary means: visual transit of the No. 4 beacon and the town wharf leading lights."

It closes:

- "Note: this plan is written for the daylight transit. **Night transits are not scheduled on this route.**"

That note is the plan's only reference to darkness. There is no night passage plan for the Weerona Channel. There is no guidance anywhere on what to do if the plotter is lost during the transit, and no abort or hold point is named in the plan.

The plan's secondary means was weaker than it appeared. The town wharf leading lights bear 195 degrees but lie 2.1 nm beyond the No. 6 beacon, and they do not resolve as two lights until the vessel is inside the channel's southern third. At the point in the transit where the plotter was lost, the secondary means was therefore the No. 4 beacon alone. As established under Environment, that beacon could not be seen from the stretch of channel the vessel then covered.

The one provision in OPS-3 that points toward slowing down or stopping is section 3.6:

- "In reduced visibility the master will reduce to a speed at which the vessel can be stopped within the distance in which the channel can be seen, and will consider anchoring or returning to open water."

It applies only in reduced visibility. On a night of more than 10 km visibility it was not engaged, even though the master had lost his means of fixing position.

Both masters interviewed said independently that the right response to losing the plotter in the channel was to stop, and both said that nothing tells a master to do so. The master's own account was that he did not think of stopping and went to the visual transit because the plan presents it as the backup. This is the gap in the procedure made visible.

A named hold point would have been available throughout the 114 seconds between the freeze and the vessel crossing the channel edge, on flat water with no other traffic. It cannot be certain

that a written rule would have been applied under the same pressure by a master who did not think of stopping unprompted. The evidence nonetheless indicates that the absence of such a rule removed the most direct defence against this sequence.

It is virtually certain that there was no night passage plan for the channel and no hold point if the primary means of fixing was lost, and very likely that this contributed to the grounding.

Change-of-master defect brief

Tarrakin Ferries Operations Manual OPS-9 governs defect reporting. Section 9.1 requires immediate reporting of:

- "A defect which renders the vessel unseaworthy or which affects a statutory item of equipment"

The chart plotter is not a statutory item on a class 1D vessel on this route, although it is the primary means of position fixing named in the passage plan. Its defects therefore fall under section 9.2:

- "All other defects shall be entered in the vessel defect book at the end of the shift in which they are found, and will be reviewed by the Operations Manager weekly."

For a fault that clears itself, an end-of-shift requirement provides no effective trigger. Section 9.3 requires the outgoing master to brief the relieving master on:

- "the state of the vessel, any defect, and anything else affecting the conduct of the service."

Nothing requires that brief to be written down. Nothing requires the relieving master to confirm that a defect mentioned in it has been entered in the defect book.

Both masters described the same understanding independently: the defect book is for what is broken at handover, and what has come good is talked about. On the day, that understanding produced the outcome the rule permits. An intermittent fault on the vessel's primary means of position fixing was known to two masters and was invisible to everyone ashore. Nobody ashore was in a position to take the vessel out of service or apply the firmware fix.

There are two readings of this factor's part in the grounding, and the evidence supports both to a degree.

- **The case that it contributed.** The point is not the single missed entry on the day. Three or four freezes over a year never reached anyone who could act on them. As a result, the firmware issued in 11/2024 was never applied, and the master met the fault on the night with no reason to regard it as more than a nuisance.
- **The case that it did not.** A defect book entry made at about 13:05 would have been reviewed by the Operations Manager at the end of that week, not that afternoon, and the vessel would have sailed TF18 either way. The path from a written entry to a different outcome on the night runs through a same-day decision that nobody in the system was required to make. On that reading, the handover rule is a genuine finding about the fleet but not a cause of this grounding.

On balance the investigation accepted the first reading. Its confidence is lower than for the more direct factors in this sequence, and the second reading has real weight.

It is virtually certain that the change-of-master defect brief was verbal, with nothing checking that what was said was written down, and likely that this contributed to the grounding.

Organisation

Winter timetable change

The current route risk assessment for the Weerona Channel is RRA-07 issue 2, dated 03/2023. It names among the hazards:

- "Boolarong Bank to the west of the channel; channel width 40 m against a vessel beam of 8.2 m; single unlit reach between the No. 2 and No. 4 beacons"

It carries the restriction:

- "Operating restriction: daylight transits only. A night transit of the Weerona Channel requires the General Manager Operations' approval and a specific passage plan."

Change record CHG-2026-014 moved TF18 from 16:15 to 17:35 for the winter timetable from 01/06/2026, to meet the afternoon coach connection. It assessed the change as:

- "timetable change only. Existing route, existing vessel, existing crew qualifications, existing berths. No new hazard identified."

It recorded:

- "Route risk assessment review required: No."

The change was approved by the Operations Manager on 22/04/2026.

The daylight-only restriction is not referred to anywhere in CHG-2026-014. As a result:

- RRA-07 was not reopened, and no issue 3 exists;
- no night passage plan was written; and
- the General Manager Operations' approval, which RRA-07 requires for a night transit, was neither sought nor given.

The change was approved one level below the role the restriction names. From 01/06/2026, every TF18 transit of the channel was made after the end of civil twilight. TF18 ran 31 times in darkness before the occurrence, and the master made 26 of those transits.

The change process asked whether anything was new rather than whether the existing assessment still held. Asked the second question, the process would have met a live restriction that the change directly contradicted.

The existence of this factor is not stated as certain. The two documents are current controlled copies. However, the Operations Manager's account of how he understood the restriction was given verbally and not in formal interview.

A review of RRA-07 against a night operation would have reached the unlit reach between the No. 2 and No. 4 beacons, which the assessment already names as a hazard. Under the assessment's own terms, it would have required a specific passage plan before the first night service ran. It cannot be certain that such a review would have produced a plan saying more than the daylight plan does. The link to the absent night passage plan and hold point is nonetheless direct.

It is extremely likely that the winter timetable moved the last service into darkness without reopening the route risk assessment, and likely that this contributed to the grounding.

Solar conversion of the No. 4 beacon

In 11/2025, the No. 4 beacon was converted from a mains supply to a solar lantern under the Council's waterways energy programme. The replacement lantern is rated at a nominal range of 3 nautical miles; the lantern it replaced was rated at 6. The character, Fl G 2.5s, did not change. The check on 18/07/2026 measured a luminous range of 2.6 nm.

A notification was lodged with the hydrographic service on 04/02/2026, and the chart correction was still pending on 07/07/2026. No local notice to mariners was issued, and no operator using the channel was told of the change.

The operator's passage plan, issue 2 and current, still listed the beacon as:

- "No. 4 (Fl G 2.5s, range 6 M)"

On the night, the master was working to a written plan that described a light twice the rated range of the one in the beacon. That lantern had been in place for about eight months.

The argument for influence is that a weaker light is harder to acquire. The master spent 91 seconds failing to acquire it. Part of that argument relies on the beacon being seen against a brighter background. Whether the new foreshore floodlighting affected detection is the matter discussed under Environment, which could not be resolved, so that part of the argument inherits the same uncertainty.

The stronger qualification is that the stretch in which the master was looking is also the stretch in which the beacon was screened entirely by the mangrove canopy. A screened light is not made visible by being brighter. For that reason this factor's influence is put materially below certainty.

It is virtually certain that the No. 4 beacon was converted to solar with its nominal range halved and no operator was told, and likely that this contributed to the grounding.

Deferral of the clearing of the beacon's sightline

The Port Authority's aids to navigation register documents the screening of the beacon in its own words, going back three years:

- **WO-118, raised 09/2023.** Recorded mangrove regrowth on Weerona Islet screening the No. 4 beacon from the northern approach between about half a mile and a quarter mile. It recommended trimming to 2 m, then on a three-yearly cycle, at an estimated \$18,500.
- **Works rounds 2023 to 2025.** The work was deferred at each of the three rounds as unfunded and referred to the next round.
- **WO-176, raised 03/2026.** Recorded the same sightline as fully closed on the northern approach.
- **Reason for the deferral.** The Authority's file note against WO-176 records that the islet is 240 m from the beacon. It therefore falls outside the 50 m vegetation scope in schedule 2.4 of the services agreement, so the work competes with boat ramps and pontoons in the general waterways vote rather than with aids to navigation.

The Council's 2019 aerial survey gave the canopy at 2.1 m. On 18/07/2026 it measured 5.8 m, against a lantern height of 4.2 m. The canopy grew past the lantern's height at some point after 2019, and the register has recorded the screening since 09/2023.

The monthly check did not detect the screening. The register records the beacon as observed and operating on the monthly check in every month from 01/2024 to 07/2026, the last on 02/07/2026. The services agreement requires:

- "S2.1 The Authority will inspect each lit aid by daylight once per quarter and will confirm each lit aid's operation by observation from the town wharf once per month."

The town wharf is 2.1 nm south of the beacon, on its far side, and Weerona Islet is not in the line of sight from there. The monthly check confirmed that the lantern was lit, from the one direction in which the screening could not be seen.

The screened arc measured on 18/07/2026 runs from 0.84 nm to 0.34 nm north of the beacon, and the vessel left the channel inside it. The only allowance against this factor's influence is that the master might have failed to pick up the light in that minute and a half even had it been visible.

It is virtually certain that clearing of the beacon's sightline was deferred unfunded for three works rounds, and very likely that this contributed to the grounding.

Aids to navigation standard on this waterway

The two port decisions above were made within an external framework. Waterways Aids to Navigation Code WAN-2 section 4.2 provides:

- "Where an aid to navigation is established on a declared shipping channel, the responsible authority must record its availability, must achieve an availability of not less than 99.0 per cent measured annually, and must notify a change to the aid's character, position or range by notice to mariners before the change takes effect."

The Weerona Channel is not a declared shipping channel, so section 4.3 applies instead:

- "Where an aid to navigation is established on a waterway that is not a declared shipping channel, section 4.2 does not apply. The responsible authority must be able to demonstrate that the aid is inspected at intervals it has determined and recorded."

The services agreement sets that interval at a quarterly daylight inspection and a monthly observation from the town wharf. Section S2.4 limits vegetation management to within 50 m of an aid. Section S2.7 states:

- "There is no requirement in this agreement to notify waterway users of a defect, a deferral or a change to an aid."

The Authority was not in breach of either instrument. No defect had ever been recorded against the No. 4 beacon, and no notice about it had ever been issued to a waterway user. The Authority held no record of any operator asking about the channel's aids, and its aids to navigation register had never been provided to Tarrakin Ferries.

The influence of this framework on the grounding was argued, and the evidence supports two readings.

- **The case that it contributed.** Every mechanism that would have carried the beacon's condition to the people using it exists in the code and stops at this channel's boundary: a measured range, a recorded availability figure, a notice to mariners. What remained was informal word of mouth between masters. The notification duty in section 4.2 is the one mechanism that would have put the range change in front of the operator, whose passage plan still carried the old figure on the night.
- **The case that it did not.** Nothing in the code stopped the Authority from notifying operators, trimming the mangroves or measuring the light. It did not do those things for reasons of money and priority, not because the code allowed it not to. An availability target of 99.0 per cent would not have detected this condition at all, because the lantern was lit and working throughout. The problem was that it could not be seen, which no availability figure measures. On that reading, the code's scope is background rather than cause.

On balance the investigation accepted the first reading, mainly on the strength of the notification duty and the out-of-date figure in the operator's plan. This finding rests on the weakest influence in this analysis. It is put below the confidence of the other contributing factors, and the contrary reading remains substantial.

The effect of the code's threshold is that a 40 m channel used by a 96-passenger ferry has beacons that nobody is required to measure, record availability for, or report on to the channel's users.

It is virtually certain that aids to navigation on this waterway carried no availability standard and no duty to inform users, and likely that this contributed to the grounding.

Hypotheses tested and excluded

Two further explanations were tested and ruled out.

Water level below prediction. This was tested because the vessel grounded about three hours into a falling tide, after two days of offshore wind. The Port Authority's gauge recorded 0.91 m at 18:12 against a predicted 0.94 m. Its largest departure from prediction in the twelve hours to 20:00 was 0.07 m. The water level was as predicted, and the vessel touched because it was 54 m outside a channel dredged to 3.2 m.

Steering or propulsion failure. This was tested because a five-degree heading change followed by a steady set out of the channel can resemble a jammed rudder or a lost engine. It was ruled out on two independent grounds:

- **Recorded parameters.** Both engines ran within 12 rpm of each other, with no alarm and no throttle movement, until 18:11:52. Steering hydraulic pressure stayed within its normal band throughout, and rudder angle followed helm demand within 1.5 degrees. The rate of turn rose and returned to zero, which is a heading applied and held rather than a sheer.
- **Post-occurrence examination.** The whole steering system was removed and function tested on 12/07/2026. Every component was serviceable, with no sign of a pre-existing defect. The damage to the starboard shaft, propeller and rudder dated from the contact.

Findings summary

These findings are the investigation's conclusions about the factors and conditions that shaped the grounding of MV Tarrakin Star on Boolarong Bank on 07/07/2026. They do not apportion blame or liability to any person or organisation.

Contributing factors

- It was very likely that the winter timetable moved the last service, TF18, into darkness without the route risk assessment being reopened, and this likely contributed to the grounding. (Safety Issue)
- With virtual certainty, the operator had no night passage plan for the Weerona Channel and no hold point for the case where the primary means of position fixing was lost, and this very likely contributed to the grounding. (Safety Issue)
- The master's decision to continue the transit at eight knots after the plotter froze, rather than stopping in the channel, was virtually certain to have occurred, and it very likely contributed to the grounding.
- It was virtually certain that shoaling was not detected until sixteen seconds before contact, when the echo sounder's shallow alarm sounded at its 0.9 m commissioning setting, and this very likely contributed to the grounding.
- The change-of-master defect brief required by the operator was, with virtual certainty, verbal, with nothing checking that what was said got written down, and this likely contributed to the grounding because intermittent plotter faults never reached anyone ashore who could act on them. (Safety Issue)
- On the day of the occurrence the plotter fault was passed on verbally at the change of master and not entered in the defect book, which was virtually certain, and this omission likely contributed to the grounding.
- It was very likely that the chart plotter display froze during the channel transit at 18:07:06 and did not recover, and the loss of the primary means of position fixing very likely contributed to the grounding.
- Aids to navigation on the Weerona Channel carried, with virtual certainty, no availability standard and no duty to tell users anything, because the channel was not a declared shipping channel, and this likely contributed to the grounding. (Safety Issue)
- The No. 4 beacon was converted to solar power, its nominal range was halved, and no operator was told, all of which was virtually certain, and this likely contributed to the grounding. (Safety Issue)
- Mangrove canopy on Weerona Islet very likely meant the No. 4 beacon was not visible from the channel over the half mile before the vessel left it, and this very likely contributed to the grounding.
- The vessel, with virtual certainty, left the dredged channel and tracked onto Boolarong Bank, and this was virtually certain to have contributed to the grounding.
- Clearing of the beacon's sightline was deferred unfunded for three works rounds, which was virtually certain, and the deferral very likely contributed to the grounding. (Safety Issue)

Other factors that increased risk

- It was virtually certain that the forward saloon lifejacket locker was not reachable from most of the saloon it served, following an interior refit that did not trigger a re-survey of

the safety equipment arrangement, and this very likely increased risk to passengers in an emergency. (Safety Issue)

- The master was, with virtual certainty, on day nine of twelve, on a split duty across a thirteen hour forty spread, and this rostering pattern for the route's winter darkness transits likely increased risk.

Other findings

- It was very unlikely that the water level was below prediction and reduced the depth over Boolarong Bank, and that hypothesis was examined and ruled out.
- The possibility that a steering or propulsion failure caused the vessel to sheer off track was very unlikely, and it was examined and ruled out.
- With virtual certainty, the master stopped the engines and made no attempt to refloat on a falling tide, and this decision very likely reduced the risk to those on board.

The investigation also examined whether new foreshore floodlighting behind the No. 4 beacon reduced the master's ability to pick up the light. The evidence did not settle that question either way. It is not a finding, and it is discussed as an unresolved matter in the Analysis.

Appendices

Appendix A - Safety factor test summary

Each factor was tested in three ways:

- **Existed:** whether the factor existed.
- **Influenced the occurrence:** whether it influenced the occurrence.
- **Importance to safety:** where it did not influence the occurrence, whether it was still important to safety.

The table uses the likelihood terms exactly as the tests produced them. In the Findings summary, "extremely likely" is expressed as "very likely", so that all findings use the same scale. A test that was not needed to settle a factor's category is shown as not tested.

Factor	Lane	Existed	Influenced the occurrence	Importance to safety	Resulting category
Aids to navigation on this waterway carry no availability standard and no duty to tell users anything	External Influences	Pass – virtually certain	Pass – likely	Not tested	Contributing factor (Safety Issue)
Beacon converted to solar, nominal range halved, and no operator was told	Organisational Influences	Pass – virtually certain	Pass – likely	Not tested	Contributing factor (Safety Issue)
Clearing of the beacon's sightline deferred unfunded for three works rounds	Organisational Influences	Pass – virtually certain	Pass – very likely	Not tested	Contributing factor (Safety Issue)
Winter timetable moved the last service into darkness without reopening the route risk assessment	Organisational Influences	Pass – extremely likely	Pass – likely	Not tested	Contributing factor (Safety Issue)
No night passage plan for the channel, and no hold point if the primary means of fixing is lost	Risk Controls	Pass – virtually certain	Pass – very likely	Not tested	Contributing factor (Safety Issue)
Change-of-master defect brief is verbal, with nothing checking that what is said gets written down	Risk Controls	Pass – virtually certain	Pass – likely	Not tested	Contributing factor (Safety Issue)
Forward saloon lifejacket locker not reachable from most of the saloon it serves	Risk Controls	Pass – virtually certain	Fail – unlikely	Pass – very likely	Other factor that increased risk (Safety Issue)
No. 4 beacon not visible from the channel over the half mile before the vessel left it	Local Conditions	Pass – extremely likely	Pass – very likely	Not tested	Contributing factor

Chart plotter display froze during the channel transit and did not recover	Local Conditions	Pass – very likely	Pass – extremely likely	Not tested	Contributing factor
Day nine of twelve, on a split duty across a thirteen hour forty spread	Local Conditions	Pass – virtually certain	Fail – unlikely	Pass – likely	Other factor that increased risk
Plotter fault passed on verbally at the change of master and not entered in the defect book	Individual Actions	Pass – virtually certain	Pass – likely	Not tested	Contributing factor
Transit continued at eight knots after the plotter froze, rather than stopping in the channel	Individual Actions	Pass – virtually certain	Pass – extremely likely	Not tested	Contributing factor
Engines stopped and no attempt made to refloat on a falling tide	Individual Actions	Pass – virtually certain	Pass – very likely (reduced risk)	Not tested	Other finding (positive)
Vessel left the dredged channel and tracked onto Boolarong Bank	Occurrence Events	Pass – virtually certain	Pass – virtually certain	Not tested	Contributing factor
Shoaling not detected until sixteen seconds before contact	Occurrence Events	Pass – virtually certain	Pass – extremely likely	Not tested	Contributing factor
Water level below prediction reduced the depth over Boolarong Bank	Local Conditions	Fail – very unlikely	Not tested	Not tested	Excluded hypothesis
Steering or propulsion failure caused the vessel to sheer off track	Occurrence Events	Fail – very unlikely	Not tested	Not tested	Excluded hypothesis
New foreshore floodlighting behind the beacon reduced the master's ability to pick it up	Local Conditions	Inconclusive – about as likely as not	Not tested	Not tested	Unresolved; not a finding

Appendix B - Excluded hypotheses

Water level below prediction reduced the depth over Boolarong Bank

Why it was examined. The vessel grounded three hours into a falling tide. An offshore wind had been blowing for two days, and a set-down of a few tenths of a metre would have mattered.

Tide gauge. The hypothesis was ruled out using the Port Authority's gauge at Kanooka Heads:

- At 18:12 the gauge recorded 0.91 m, against a prediction of 0.94 m, a difference of 0.03 m.
- In the twelve hours to 20:00, the largest departure from prediction was 0.07 m.
- The gauge and the prediction are independent of each other and of the vessel.

Depth at the contact point. The bed was surveyed at 0.50 m below chart datum, so there was 1.41 m of water over it. The vessel's static draught was 1.35 m, and its calculated squat at 6.8 kt was 0.18 m.

Conclusion. The vessel touched because it was 54 m outside a channel dredged to 3.2 m, not because the water was lower than the chart and tide table predicted.

Steering or propulsion failure caused the vessel to sheer off track

Why it was examined. From outside, a five-degree heading change followed by a steady set out of the channel is what a jammed rudder or a lost engine would look like.

The hypothesis was ruled out on two independent grounds.

Recorded data. From 18:03 to 18:11:52:

- Both engines ran at 1,780 and 1,784 rpm, within 12 rpm of each other, with no alarm, no excursion and no throttle movement.
- Steering gear hydraulic pressure stayed between 96 and 99 bar, against a normal band of 90 to 105 bar.
- Rudder angle followed helm demand within 1.5 degrees throughout.
- The rate of turn peaked at 2.1 degrees per minute and returned to zero. This is consistent with a heading being applied and then held, not with a sheer.

Physical examination. On 12/07/2026 the whole steering system was removed and function tested. Every component was serviceable, with no internal leakage and no sign of a pre-existing defect. The damage to the starboard shaft, propeller and rudder all dated from the contact, not before it.

Bridge audio. The bridge audio contains no remark about steering or engines at any point.

Appendix C - Sources of evidence

Interviews and accounts

Person (role)	Date	Method
The master	14/07/2026	Interviewed and recorded
The deckhand	14/07/2026	Interviewed; not recorded, at her request
The outgoing (morning) master	23/07/2026	Interviewed by telephone; not recorded. The notes were read back to him and agreed.
The Operations Manager	[NOT ESTABLISHED: date]	Gave a verbal account of how he understood the daylight-only restriction; not taken as a formal interview

Safety reports

- SR-2026-0288, the master's safety report, timed 18:52 on 07/07/2026.
- SR-2026-0291, the deckhand's safety report.

Recorded data and audio

- **AIS-2026-0093-R1:** Port Control AIS receiver log and vessel recorder readout, issued 15/07/2026. Covers the AIS track, engine monitoring recorder and steering data for service TF18.
- **AUD-2026-0093:** VHF channels 12 and 16 and bridge camera audio, 17:32 to 19:26 local. Transcribed 16/07/2026 and verified by the Port Authority's traffic officer. The bridge camera was downloaded before the vessel was slipped.

- **Chart plotter internal event log:** recovered 09/07/2026, with the contractor's bench test report of 21/07/2026.
- **Echo sounder setup and alarm log:** recovered 09/07/2026.
- **MET-TIDE-0093-070726:** tide predictions and gauge record, weather observations, and light and moon data, retrieved 09/07/2026.

Surveys, examinations and measurements

- Grounding position survey and bed levels, surveyed 08/07/2026.
- Post-grounding survey, slipping and steering gear examination, MV Tarrakin Star, 09/07/2026 to 14/07/2026. The steering system function test was conducted on 12/07/2026.
- Forward saloon lifejacket stowage survey, MV Tarrakin Star, 16/07/2026.
- Range and visibility check of the No. 4 beacon, 18/07/2026. This was made from the port workboat, on the inbound track, at the same time of night as the occurrence.
- Kanooka Heads foreshore lighting survey, 18/07/2026, made from the port workboat.

Operator documents

- Tarrakin Ferries Operations Manual OPS-3, revision 6 (02/2023), passage planning, with appendix A issue 2, the Weerona Channel plan.
- Tarrakin Ferries Operations Manual OPS-9, revision 4 (08/2024), defect reporting and rectification.
- Route Risk Assessment RRA-07, issue 2 (03/2023), and change record CHG-2026-014, approved 22/04/2026.
- Crew records, certificates and the roster for 07/07/2026, copied 13/07/2026.
- Vessel maintenance history and wheelhouse defect book, MV Tarrakin Star, copied 10/07/2026.

Port Authority, Council and regulatory documents

- Kanooka Heads Port Authority aids to navigation register and register extract for the No. 4 beacon, copied 19/07/2026. These include work orders WO-118 and WO-176.
- Aids to navigation services agreement between Kanooka Shire Council and Kanooka Heads Port Authority, schedule 2.
- Waterways Aids to Navigation Code WAN-2, section 4.

Appendix D - Investigation activity

07/07/2026

- The investigation was notified by the master's safety report SR-2026-0288, timed 18:52, before the operator's notification was raised.
- The occurrence was assessed as a serious marine incident warranting investigation, and the investigation was opened.

08/07/2026

- The vessel had been refloated overnight and was secured at the town wharf.
- The chart plotter and echo sounder heads were removed and sealed.
- The Port Control AIS log and VHF recordings were requested, and the bridge camera was downloaded before the vessel was slipped.
- The master and the deckhand were stood down and tested; the results were negative.

- The General Manager Operations suspended the evening Muntari services.

10/07/2026

- The grounding position survey and bed levels were received; the survey had been run on 08/07/2026.
- The steering gear was removed for function testing.
- The deckhand's safety report SR-2026-0291 was received, and the forward saloon lifejacket stowage was added to the scope of the investigation.

14/07/2026

- The master was interviewed and the interview recorded. The deckhand was interviewed but not recorded, at her request. Both raised the No. 4 beacon.
- A night range and visibility check of the beacon was requested, to be made from the port workboat on the inbound track.

19/07/2026

- The range and visibility check, made on 18/07/2026, was completed. It found that the No. 4 beacon was not visible from the channel between 0.84 nm and 0.34 nm north of it.
- It was established that the solar conversion of 11/2025 had halved the beacon's nominal range, with no notice given to any operator.
- The foreshore lighting was surveyed the same night.
- The Port Authority's aids to navigation register was obtained, showing that work order WO-118 had been deferred at three works rounds.

23/07/2026

- The outgoing master was interviewed by telephone. The interview was not recorded; the notes were read back to him and agreed.
- Both masters interviewed had raised the beacon unprompted, and neither had ever reported it.
- The company's 'navigation hazard' reporting category was found to be unused on this route and unknown to the masters.
- The beacon was raised with the Port Authority and the operator.

Distribution

Title	Name
General Manager Operations	H. Vardon
Safety Manager	P. Nandasiri
Operations Manager	C. Bickerton
Harbour Master, Kanooka Heads	L. Truscott
Aids to Navigation Officer, Kanooka Shire Council	E. Maguire